

Selective Mitochondrial-Redox Catastrophe in Cancer Cells via Sub-Electroporation Eddy Currents from Intermediate-Frequency Alternating Magnetic Fields (LF-EMF)

A mechanistic hypothesis and experimental design proposal for a non-thermal EMF monotherapy

Derived exclusively from the 49-paper / 434-condition non-thermal EMF dataset analysis

1. Rationale and scope

Among the six candidate non-thermal field types, intermediate-frequency alternating magnetic fields (AMF) — the report's macro F, labelled LF-EMF, spanning 87–320 kHz at 1.4–18 mT — provide the strongest joint mechanistic-and-selectivity rationale. Four discriminating features, all documented in the report, drive the choice.

First, selectivity. Macro F carries the highest interpretable selectivity index in the dataset (SI = 11.1, from a 56% cancer therapeutic rate against a 0% healthy-harm rate over $n = 11$ healthy conditions), and it is the only category in which a single matched experiment exposes nine cancer lines (U87, A172, MCF-7, MDA-MB-231, MDA-MB-453, LN229, U251, T98, PANC-1; viability reduced to 48–88% of control) against three healthy lines (NHA, HCF, HUVEC; 90–105%, all Direction = 0) under identical parameters (Paper 4: 227 kHz, 18 mT, 30 min).

Second, dose structure. Macro F is among the few categories with an internal dose–response: a frequency screen on U87/LN229 localises maximal sensitivity to 198–227 kHz with effects fading at the band edges (320 kHz → 92%, Direction = 0), and an amplitude gradient across the three contributing papers establishes a threshold (weak at ≈ 1.5 mT; large at 10–18 mT). A tunable, falsifiable parameter window is therefore already defined.

Third, deliverability. A magnetic field couples to tissue without electrodes and penetrates depth, unlike the contact-dependent, surface-limited electric route. This is the report's own argument for prioritising the magnetic over the electric realisation of the same 100–300 kHz band.

Fourth, the competitors are disqualified on the report's evidence. ELF-EMF (macro K) is non-selective at cytotoxic amplitudes (37% of healthy measurements affected; hormetic pro-proliferative inversion at low amplitude) and is graded adjuvant-only; SMF (H) rests on 4 conditions / 2 papers with an unstable proliferation sign; RF-EMF (G) has zero healthy controls, a counter-therapeutic ROS reduction, and frank pro-proliferative windows; PMF (B) is single-paper, single-cell-line (C6), viability-only, with zero healthy controls; nsPEF (A) achieves selectivity only on one isogenic-style pair at one pulse duration and operates in a contact-dependent intracellular-electroporation regime. Macro F alone is simultaneously selective, dose-resolved, mechanistically coherent, and clinically deliverable.

Scope is confined to the sub-electroporation regime of macro F (sinusoidal AMF, 198–227 kHz, 5–18 mT, 30–60 min). The high-amplitude pulsed and static-field regimes, which the report assigns to distinct biophysical categories, are out of scope.

2. Mechanistic hypothesis

Core statement. A sinusoidal alternating magnetic field at 198–227 kHz and 5–18 mT induces intracellular eddy currents whose dissipative coupling to the mitochondrial compartment is tolerated by healthy cells but, in cancer cells bearing a Warburg-type mitochondrial and redox phenotype, precipitates a self-amplifying redox catastrophe that terminates in non-apoptotic loss of viability through cell-cycle arrest rather than programmed death. The chain has four connected axes: one biophysical, three biological.

2.1 Primary transduction: a sub-electroporation induced electric field

A uniform sinusoidal B-field induces inside a cell of radius r an azimuthal electric field of order $E_{\text{ind}} \approx \pi f B r$; at 227 kHz, 18 mT, $r \approx 8 \mu\text{m}$ this is ≈ 0.1 V/m, three to four orders below the kV/cm electroporation threshold of nsPEF. The induced current density scales as $f \times B \times r^2$, which explains — without invoking heating — both the frequency optimum (≈ 200 –230 kHz) and the amplitude threshold (10–18 mT) the report observes: efficacy

tracks the product that sets induced-current magnitude. *[Inferred biophysical step: the report computes E_{ind} but does not measure intracellular current; the dose–response it predicts is, however, directly evidenced.]*

2.2 Selective mitochondrial perturbation (the selectivity-determining step)

The induced current preferentially perturbs the inner mitochondrial membrane of cells whose respiratory chain operates near its electrochemical ceiling. Warburg-phenotype cancer cells carry elevated baseline $\Delta\Psi_m$ and a dysregulated electron-transport chain; the report attributes macro-F selectivity to exactly this differential. The proximal consequence is perturbed electron flow and a ROS pulse. **Partly evidenced:** ROS Direction = +1 in 2 of 3 cancer conditions, with metabolic-reprogramming and glycolytic-enzyme (LDH-A, HK2, PKM2) and PGC-1 α modulation in Paper 4. **Inferred and explicitly open:** the $\Delta\Psi_m$ step is the weakest link — macro F measured $\Delta\Psi_m$ in only one condition (Direction = +1, not collapse), so the direction and timing of the potential change are unresolved by the report and are designated here as the primary experimental target, not an assumption. The $\Delta\Psi_m$ -collapse motif is robust only in nsPEF (macro A, 21/21 Direction = -1), a different field; importing it into macro F would be unsupported, and this hypothesis avoids doing so.

2.3 Antioxidant-buffer collapse and oxidative lesion

The ROS pulse meets a cancer-specific vulnerability: a constitutively depleted antioxidant reserve. Macro F provides the dataset's only substantive antioxidant comparison — Antioxidant Direction = -1 dominant in cancer cells (Paper 4, 9 conditions), consistent with GSH depletion or antioxidant-enzyme down-regulation. The hypothesis: ROS production exceeds an already-eroded buffer, yielding oxidative macromolecular damage, for which the macro-F genomic block (γ -H2AX foci, comet-assay endpoints; 8 conditions) is the predicted signature. Healthy cells, with intact GSH and antioxidant enzymes, absorb the same induced-current-driven ROS and accrue no lesion — the biochemical basis of the observed 0/11 healthy-harm rate. *[Most directly evidenced axis; also the one offering a decisive causal test — see 4.4 rescue arm.]*

2.4 Metabolic arrest and a non-apoptotic cell fate

The terminal axis is cell-cycle arrest, not programmed death. Apoptosis was uniformly Direction = 0 across all four cancer and four healthy conditions measured in macro F — a strong negative the hypothesis takes at face value. The evidenced terminal signature is G0/G1 arrest (Paper 38, phase-resolved in SH-SY5Y) with cyclin D1 down and p21 up (Paper 4 proteomics): the canonical mitochondrial-stress $\rightarrow$ p21 $\rightarrow$ cyclin-D1 $\rightarrow$ G1/S-block axis the report flags as cross-cutting. **Inferred:** the causal ordering (redox lesion upstream of arrest) is not temporally resolved, and the death modality downstream of arrest is unmeasured in macro F. Because antioxidant depletion plus a ROS pulse is the canonical precondition for ferroptosis — the one regulated-death modality the report finds positive elsewhere (macro I, Paper 9) yet never tests in macro F — ferroptosis is nominated as the leading candidate terminal modality, to be adjudicated experimentally, not assumed.

Evidential status. Directly evidenced: viability selectivity, antioxidant depletion, ROS rise, γ -H2AX, G0/G1 arrest with cyclin D1 $\downarrow$ /p21 $\uparrow$, the non-apoptotic route, and the frequency/amplitude dose structure. Inferred: the induced-current biophysics (2.1), the $\Delta\Psi_m$ direction and timing (2.2), the upstream-to-downstream ordering (2.4), and the specific terminal death modality (2.4). The design below targets every inferred link.

3. Parameter selection strategy

Frequency. Test 100, 160, 198, 227, 280 and 320 kHz, centred on the report's resolved optimum (198–227 kHz) and bracketing both edges where it records loss of effect. Rationale is biophysical: induced current rises with f , but the fade at ≥ 280 –320 kHz indicates an upper bound — plausibly dielectric-relaxation or membrane-time-constant capping of effective coupling — so the band must be mapped, not assumed monotonic.

Intensity. Test 1.5, 5, 10 and 18 mT. The report fixes ≈ 1.5 mT as sub-threshold and 10–18 mT as strongly effective; 5 mT probes the inflection. Amplitude is the dominant efficacy predictor in this band and is sampled most densely. Low amplitudes also guard against the hormetic pro-proliferative inversion the report documents at low-field exposures (a risk to exclude, not assume absent).

Waveform. Pure sinusoid only, matching every macro-F study; this isolates the continuous-coupling regime from the pulsed/square regimes the report assigns to separate categories and keeps the induced-current spectrum monochromatic.

Duration and schedule. 30 and 60 min per session (the macro-F range), once daily over 5 days, with readouts at 0, 6 and 24 h post-exposure to resolve the redox-then-arrest ordering the report cannot. The 30-min point anchors to Paper 4's strongest condition; 60 min addresses the report's stated mitigation for selectivity loss (lower amplitude, longer time).

4. Experimental design

4.1 Cell lines

Class	Lines (all report-tested in macro F unless noted)	Justification
GBM (cancer)	U87, U251, LN229	Most-tested macro-F indication; high mitochondrial-respiratory dependence; post-mitotic neural surround intrinsically sparing-friendly.
TNBC (cancer)	MDA-MB-231, MDA-MB-453	Second broadly responsive lineage across the dataset.
PDAC (cancer)	PANC-1	Extends the panel to a third tumour origin tested in Paper 4.
Healthy	NHA, HCF, HUVEC, HDFa	The four macro-F healthy lines; NHA tissue-matched to GBM. HDFa is the mandatory proliferating control.
Healthy	MCF-10A (added)	Breast-matched non-tumorigenic control used elsewhere in the report for selectivity pairing.

At least one rapidly proliferating healthy line (HDFa) is mandatory to dissociate the proposed selectivity mechanism (mitochondrial/redox phenotype) from a trivial mitotic-fraction confound — the report's primary stated risk.

4.2 Exposure apparatus and non-thermal isolation

A water-cooled solenoid or Helmholtz pair (the macro-F device class) driven by a tunable RF amplifier delivering a calibrated sinusoidal B-field, 100–320 kHz, up to 18 mT, with homogeneity mapped across the vessel (<10% variation). Non-thermal isolation requires four parallel controls: (i) an in-well fibre-optic temperature probe with feedback holding $\Delta T < 1^\circ\text{C}$ (the report's non-thermal criterion); (ii) a sham-coil arm (identical coil and thermal load, field nulled); (iii) a temperature-matched bath arm reproducing any residual ΔT without field; and (iv) incubator-only controls. The sham and temperature-matched arms together separate non-thermal field action from coil-associated heating or vibration.

4.3 Outcomes (three tiers, aligned to the hypothesis axes)

Cell-fate: viability (primary; matched dose grid), proliferation (EdU, Ki-67), apoptosis (Annexin-V/PI, cleaved caspase-3 — predicted Direction = 0, confirming the non-apoptotic route), regulated necrosis (LDH release, PI uptake), senescence (SA- β -gal — a report-flagged coverage gap), and ferroptosis (C11-BODIPY lipid peroxidation, malondialdehyde — the nominated terminal modality, a macro-F gap).

Mechanistic: $\Delta\Psi_m$ by JC-1, time-resolved at 0/6/24 h (the designated primary mechanistic readout, resolving the report's single-condition gap); total and mitochondrial ROS (DCFH-DA, MitoSOX); GSH/GSSG ratio (the evidenced antioxidant axis); cell-cycle distribution (PI flow, expecting G0/G1 accumulation); and bioenergetic flux (Seahorse OCR/ECAR, operationalising the metabolic-reprogramming and glycolytic-enzyme findings).

Molecular: γ -H2AX foci (oxidative-lesion marker); cyclin D1 and p21 (arrest axis); LDH-A, HK2, PKM2 and PGC-1 α (Warburg/biogenesis axis).

4.4 Controls and selectivity test

Every cancer and healthy line is exposed to the identical (f, B, duration) grid at matched passage and confluence, paired on the same coil on the same day. The selectivity index is pre-registered as (cancer viability reduction)/(healthy viability reduction) at matched parameters, with the report's SI ≥ 4 as success. The decisive causal control is an **antioxidant-rescue arm**: co-treatment with N-acetylcysteine (global ROS) and MitoTEMPO (mitochondria-targeted ROS) should abolish cancer-cell cytotoxicity if axes 2.2–2.3 are causal — directly addressing the report's statement that the ROS/antioxidant-to-viability link has never been shown by rescue. To counter single-paper anchoring, the 227 kHz / 18 mT / 30 min condition is replicated across two independent apparatus builds and operators under blinded analysis.

5. Predictions and falsification

If the hypothesis holds, the following pattern is expected:

- **Selectivity.** Cancer viability falls $\geq 30\%$ (proliferation $\geq 50\%$) at 198–227 kHz, ≥ 10 mT, 30–60 min; matched healthy lines stay at Direction = 0 (SI ≥ 4), including the proliferating control — excluding the mitotic-fraction confound.
- **Frequency window.** Effect peaks at 198–227 kHz and is lost at 100 kHz and ≥ 320 kHz (non-monotonic band), confirming an induced-current window rather than a monotonic dose.
- **Amplitude threshold.** Effect is weak/absent ≤ 1.5 mT, inflects near 5 mT, and saturates by 18 mT; any low-amplitude viability increase (hormesis) is mapped but does not falsify.
- **Redox causality.** In cancer only, ROS $\uparrow$ and GSH/GSSG $\downarrow$ precede viability loss, and NAC + MitoTEMPO abolish the cytotoxicity.
- **Death route.** Caspase-3/Annexin-V remain unchanged (non-apoptotic confirmed); positive C11-BODIPY/MDA (with GSH $\downarrow$ and $\Delta\Psi_m$ change) would identify ferroptosis, positive SA- β -gal senescence — the design adjudicates between them.
- **Arrest signature.** G0/G1 fraction rises with cyclin D1 $\downarrow$ / p21 $\uparrow$ and γ -H2AX $\uparrow$ in cancer, minimal in healthy.
- **$\Delta\Psi_m$.** The discriminating readout: an early cancer-specific $\Delta\Psi_m$ perturbation (direction and timing to be established) coupled to the ROS rise, with healthy $\Delta\Psi_m$ stable.

Falsification. The hypothesis fails if any of the following hold: (i) healthy lines lose viability comparably to cancer at the selective window (SI < 2), collapsing selectivity; (ii) antioxidant rescue does not attenuate cancer cytotoxicity, severing the redox chain; (iii) cytotoxicity is independent of baseline $\Delta\Psi_m$ / Warburg status across the panel, refuting the mitochondrial-vulnerability basis of selectivity; (iv) the dose–response shows neither a frequency optimum nor an amplitude threshold, refuting the induced-eddy-current transduction model. Each falsifier maps, by construction, to a specific inferred link in Section 2.